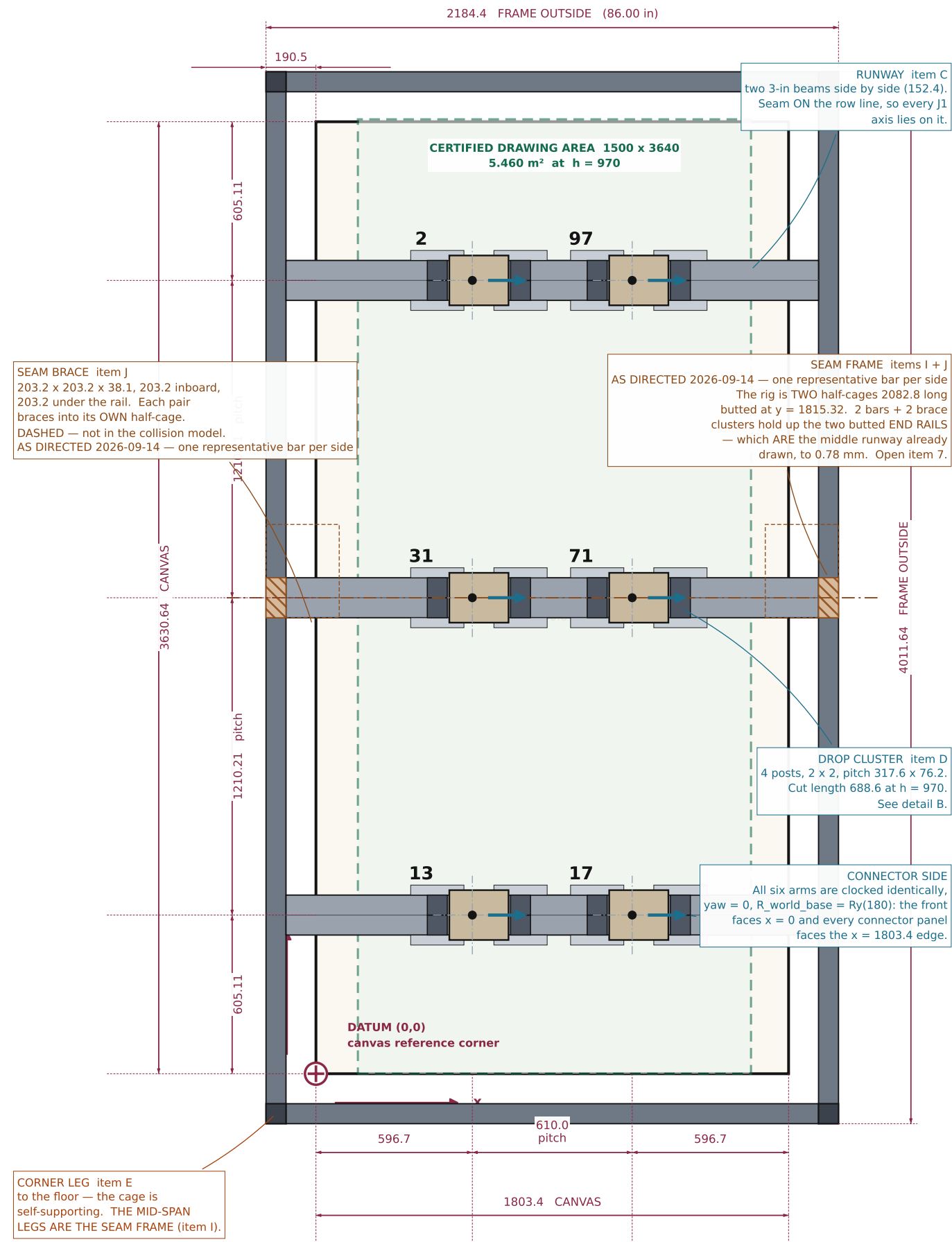


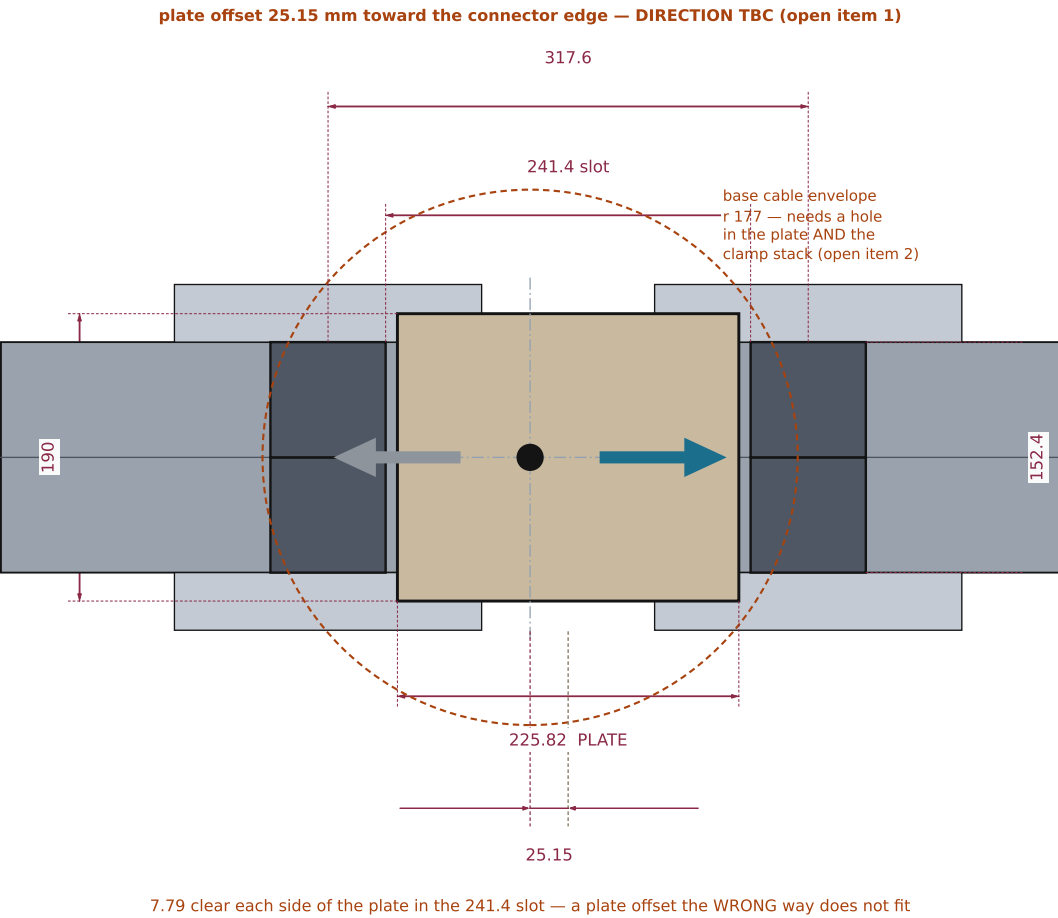
A PLAN — BASE SPACING AND THE STEEL OVER IT

viewed from below (arms hang toward the reader) · scale 1 : 20



B DETAIL — ONE DROP CLUSTER IN PLAN

at the J1 axis · scale 1 : 5



C SET-OUT AND WHAT THE SPACING BUYS

Base positions are to the J1 AXIS — the centre of the base bolt circle — not to a plate edge.

columns $x = 596.7 / 1206.7$ pitch 610.0
rows $y = 605.11 / 1815.32 / 3025.53$ pitch 1210.21
Tolerance +/-10 mm per base; record any as-built offset rather than re-centring the others.

ORIENTATION — load-bearing, and identical for all six.
 $R_{world_base} = Ry(180)$, yaw = 0. The flipped base's +x (the arm's front) points toward x = 0, so every connector panel faces the x = 1803.4 long edge. Do not clock any arm differently: the certified coverage and every program assume this exact uniform orientation.

CERTIFIED DRAWING AREA at h = 970: 1500 x 3640 mm, 5.460 m².
x 160.1640, y 0.3620 from the datum corner. 97.72 % of the canvas is live, but only this rectangle is hole-free. Read at run time from out/certified_area_h0970.json and drawn dashed on panel A, so the spacing can be read against what it actually buys.

THE STEEL DOES NOT FIT THE CERTIFIED KEEP-OUT.
Every certified number in this repo was earned against a single 200 x 200 mm column on the base axis. The real cluster is 393.8 x 152.4 offset 25.15 mm off it, plus 203.2 mm gussets: all 60 pieces of mount hardware fall outside it, worst 185.55 mm. Re-certification is minutes of compute and it has not been run — open item 6.

NOTES

1 DATUM

z = 0 is the TOP SURFACE OF THE PAPER as laid on the table. The floor is -636.68 mm; the paper therefore sits 636.68 mm above it. x across the short side (0 -> 1803.4), y along the long side (0 -> 3630.64), z up.

2 THE HEIGHT ON THIS SHEET

h = 970.0 mm. Drop post 688.6 mm.

Alternates, same sheet, same steel:

h = 970 post 688.6 (this sheet)

h = 940 post 718.6 (what ships)

h = 850 post 808.6 (built today)

Only item D changes with h.

3 MEMBER SECTIONS

3" x 3" (76.2 mm) square T-slot throughout, the original drawing's own MTEXT. Gussets 1.5" x 3" stock, 203.2 x 203.2.

4 THE RUNWAY SEAM

Each row is TWO 3-in beams side by side (152.4 overall). The seam lands ON the row line, so each arm's J1 axis lies on it and the 2 x 2 cluster straddles it.

4b THE SEAM FRAME — items I + J

AS DIRECTED 2026-09-14 — one representative bar per side

The rig is TWO half-cages 2082.8 long butted at y = 1815.32. Their two seam-side END RAILS *ARE* the middle runway already drawn (0.78 mm) — do not cut them twice. 2 bars + 2 brace clusters hold them up.

5 GUSSETS ARE ROTATED

The drawing's inboard orientation needs 406.4 mm across a transverse pair and only 216.2 exists. Rotated onto the runway's OUTBOARD y faces they clear by 89.2 mm and brace the cluster's weak axis (the posts are only 76.2 apart in y).

6 KEEP-OUT BELOW THE MOUNT PLANE

Nothing but the six arms below z = 970 over the canvas plus 1 m all round. No braces, no cable drops, no lights. The posts already run 34.98 mm past the plate underside.

7 OPEN ITEMS THAT BLOCK CUTTING

1 PLATE OFFSET DIRECTION

2 BASE CABLE PASS-THROUGH

3 CAGE LEGS

4 GUSSET PART AND ATTACHMENT

6 RE-CERTIFICATION OF THE DROP CLUSTER

Full text in out/drawings/8020_cut_list.md.

8 WHAT THIS SHEET IS NOT

NOTHING HERE IS A SURVEY.

The room has never been measured. Every ceiling number descends from the original drawing's 233,7 cm, which is FLOOR to top of a self-supporting cage.

REV A — SUPERSEDES THE 1435 / 1525 DROP POSTS of out/ceiling_8020_layout.* — that grid datum read the drawing's 233,7 cm (FLOOR to top of cage) as if it were measured from the paper. DO NOT CUT FROM THE OLD SHEET. SEAM FRAME items I + J — the two half-cages butt at y = 1815.32 and the steel that holds their two butted end rails up is NEW ON THIS REVISION: AS DIRECTED 2026-09-14 — one representative bar per side.

ARIS · six inverted FR3 over a horizontal paper table	DESIGN MOUNT PLANE h	DROP POST item D	DATUM	DATE
ARM SPACING — PLAN	970.0 mm	688.6 mm	z = 0 paper top	2026-09-16
sheet 1 of 4 · A3 · all dimensions mm · scales as noted	above the paper top surface	= (1623.62 - h) + 34.98, 24 off	floor -636.68 · runway underside 1623.62	scripts/draw_8020.py · read-only against the package